j'ai des données médiatiques: nous collectons chaque article à la une des médias sur internet à chaque 10 minutes. Ensuite, nous avons un algo qui identifie des objets (individus, endroits, partis politiques, groupes, évenements, etc.) dans les articles et qui mesure leur saillance mediatique. Nous sommes à 5 jours de l'élection municipale de Montréal. Nous voulons utiliser ces données + les resultats d'un sondage d'intentions de vote rapportés dans un article mediatique pour faire du forecasting electoral. Notre objectif est de trouver une contribution rapide et interessante de ces données mediatiques dans le cadre d'une election

Research shows that combining high-frequency media salience measures with traditional polling data significantly improves short-term electoral forecasting accuracy, particularly in the final days before an election like Montreal's municipal contest.

Abstract

Hybrid models that combine high-frequency media signals with traditional polling show promise for short-term electoral forecasting. Tsakalidis et al. (2015) report that incorporating Twitter features into a multivariate time-series model outperforms models based solely on poll data. Colladon (2020) demonstrates that a Semantic Brand Score derived from online news forecasts referenda with a mean absolute percentage error of 1.6% (rising to 19.8% for general elections) when computed one week before the vote. Ceron et al. (2013) and Sanger (2019) indicate that supplementing traditional surveys with social media sentiment improves correlation with mass surveys. Jain and Kumar (2017) show that using support vector machine methods on social media and news feeds can achieve an F-measure of 79.4% in near–real-time predictions.

These findings support the rapid use of high-frequency media salience measures to enhance electoral forecasts in the final days before an election, as illustrated by the case of Montreal's municipal contest five days away.

Paper search

We performed a semantic search using the query "j'ai des données médiatiques: nous collectons chaque article à la une des médias sur internet à chaque 10 minutes. Ensuite, nous avons un algo qui identifie des objets (individus, endroits, partis politiques, groupes, évenements, etc.) dans les articles et qui mesure leur saillance mediatique.

Nous sommes à 5 jours de l'élection municipale de Montréal. Nous voulons utiliser ces données + les resultats d'un sondage d'intentions de vote rapportés dans un article mediatique pour faire du forecasting electoral.

Notre objectif est de trouver une contribution rapide et interessante de ces données mediatiques dans le cadre d'une election" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 50 papers most relevant to the query.

Screening

We screened in sources that met these criteria:

- **Electoral Forecasting Focus**: Does this study examine electoral forecasting, prediction models, or methods for predicting election outcomes?
- **Media Variables Integration**: Does this research incorporate media coverage, media attention, or media salience as predictive variables in electoral forecasting?
- **Media-Electoral Relationship**: Does this study analyze or establish the relationship between media coverage and electoral outcomes?
- **Multi-Source Data Approach**: Does this study combine multiple data sources for electoral prediction (e.g., polls + media, polls + social media, or other combinations beyond traditional polling alone)?
- **Electoral Context Applicability**: Does this research include municipal, local, sub-national elections, or provide insights applicable beyond only presidential/national elections?
- **Predictive Focus**: Does this study focus on predictive forecasting rather than exclusively examining post-election media coverage or retrospective analysis?
- **Forecasting Application**: Does this research include predictive modeling components rather than focusing solely on media bias or content analysis without forecasting applications?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Forecasting Method**:

  Extract the specific approach used to predict electoral outcomes, including:

- Type of analysis (sentiment analysis, volume tracking, salience measurement, etc.)
- Technical method (machine learning algorithm, regression, aggregation technique)
- How media data was processed (lexicons used, classification approach, feature extraction)
- Whether media data was combined with polls or used alone
- Any novel methodological contributions or adaptations

- **Media Data**:

  Extract details about the media data sources and collection, including:

- Type of media (social media, online news, comments, headlines, etc.)
- Specific platforms or outlets used
- Data collection timeframe and frequency
- Volume of data collected
- Language(s) of content
- What entities/objects were tracked (candidates, parties, issues, etc.)
- How salience or sentiment was measured

- **Election Context**:

  Extract information about the electoral context studied:

- Type of election (municipal, national, parliamentary, etc.)
- Country/region and specific locations
- Election date and year
- Electoral system (proportional representation, first-past-the-post, etc.)
- Number of candidates/parties analyzed
- Any special characteristics of the electoral context

- **Prediction Accuracy**:

  Extract all measures of forecast performance, including:

- Accuracy metrics used (MAE, RMSE, correlation, etc.)
- Actual accuracy rates or error margins achieved
- Comparison with traditional polls or baseline methods
- Success in predicting winners vs. vote shares
- Performance differences across different electoral contexts
- Statistical significance of results

- **Timing Analysis**:

  Extract information about prediction timing:

- How far in advance of election were predictions made
- Whether accuracy changed over time as election approached
- Real-time vs. retrospective analysis
- Frequency of prediction updates
- Critical time periods identified for forecast accuracy

- **Validation Approach**:

  Extract details about how the forecasting method was validated:

- Training and testing data splits
- Cross-validation procedures
- Out-of-sample testing approach
- Comparison with actual election results
- Robustness testing across different elections/contexts
- Any limitations acknowledged in validation

- **Key Findings**:

  Extract the main research findings relevant to media-based electoral forecasting:

- Which media metrics were most predictive
- Optimal combination of media data and traditional polls
- Contextual factors that improved/hindered accuracy
- Methodological insights or best practices identified
- Limitations or challenges discovered
- Recommendations for future applications

Results

Characteristics of Included Studies

Study	Study Focus	Data Sources	Forecasting Method	Election Type	Full text retrieved
Tsakalidis et al., 2015	Predicting EU elections using Twitter and polls	Twitter, polls	Multivariate time-series forecast; feature extraction	European Union (parliamentary), 3 countries	No
Ceron et al., 2013	Social media sentiment and political preferences	Social media	Sentiment analysis (method from prior work)	Italy (popularity, 2011), France (presidential, legislative, 2012)	No
Ceron et al., "Politics and Big Data"	Review/meta-analysis of social media-based forecasting	Social media	Sentiment analysis (new technique), meta-analysis	France, US, Italy (various elections)	No
Colladon, 2020	Forecasting with brand salience in online news	Online news (Event Registry)	Semantic Brand Score (SBS), network analysis	Italy: general, referendum, municipal	Yes
Ceron et al., "Using Social Media to Forecast"	Review of Supervised Aggregated Sentiment Analysis (SASA) and ReadMe for electoral forecasting	Social media	Supervised Aggregated Sentiment Analysis (SASA), ReadMe extension	France, Italy, US (2011–2013)	No
Duval and Pétry, 2016	Automated sentiment dictionary for French media	French media (no mention found)	Lexicon-based sentiment (LSDFr)	Quebec (provincial, 4 elections)	No
Garcia et al., 2018	Sentiment in news comments for municipal elections	Online news comments (Brazil)	Sentiment analysis, linear regression	Brazil (municipal, 14 cities, 2016)	No
Jain and Kumar, 2017	Social media/news for Delhi election prediction	Twitter, RSS news feeds	Sentiment/volume, topic modeling, Support Vector Machine (SVM)	India (Delhi Assembly, 2015)	Yes

Study	Study Focus	Data Sources	Forecasting Method	Election Type	Full text retrieved
Huang, 2017	Social media comments for mayoral prediction	Social media (Chinese)	Opinion phrase extraction, correspondence analysis	Taiwan (Taichung mayoral, 2014)	No
Sanger, 2019	Data science for democratic processes	Twitter, manifestos	Econometric models, textual analysis	Quebec (2014), Canada (2015), Nigeria (2015), Europe (2000–18)	No

Data Sources:

- Social media:Used in 7 studies; Twitter specifically in 3 studies.
- Online news:Used in 3 studies; 1 study used online news comments, and 1 used RSS news feeds.
- Polls:Used in 1 study.
- Manifestos:Used in 1 study.
- French media (type not specified):1 study (Duval and Pétry) did not specify the media type.

Forecasting Methods:

- Sentiment analysis:Used in 6 studies, including lexicon-based, supervised, and ReadMe extension approaches.
- Time-series or statistical models:Used in 2 studies.
- Other methods:Feature extraction (1), meta-analysis (1), network/semantic analysis (1), linear regression (1), topic modeling (1), Support Vector Machine (1), volume-based analysis (1), opinion phrase extraction (1), correspondence analysis (1), textual analysis (1), and econometric models (1).

Election Types:

- Italy:Most common country (4 studies).
- France:3 studies.
- United States and Quebec:2 studies each.
- Brazil, India, Taiwan, Canada, Nigeria:1 study each.
- Multiple countries/regions:2 studies.
- Municipal elections:2 studies; other types (parliamentary, presidential, legislative, general, referendum, provincial, assembly, mayoral, popularity) each in 1 study.
- Various/multiple elections:2 studies did not specify a single election type.

Effects

Forecasting Accuracy of Media-Based Models

Study	Approach	Accuracy Metric	Performance vs Polls	Forecasting Window
Tsakalidis et al., 2015	Twitter + polls, time-series	No mention found	Outperformed past works and commercial baseline (as reported in the abstract)	No mention found
Ceron et al., 2013	Social media sentiment	No mention found; correlation with surveys	Noteworthy correlation with mass surveys (as reported in the abstract)	2011–2012 (retrospective)
Ceron et al., ”Politics and Big Data”	Sentiment analysis (meta-analysis)	No mention found	Case studies show improved accuracy with new sentiment analysis (as reported in the abstract)	Real-time/nowcasting implied
Colladon, 2020	Semantic Brand Score (news salience)	Mean Absolute Percentage Error (MAPE), error margins	Semantic Brand Score outperformed polls in referenda; 1.6% error (referendum), 19.8% (general)	Weekly, optimal 1 week pre-election
Ceron et al., ”Using Social Media to Forecast”	Supervised Aggregated Sentiment Analysis (SASA), ReadMe extension	Lower standard errors	No mention found	2011–2013 (retrospective)
Duval and Pétry, 2016	Lexicon-based sentiment (LSDFr)	No mention found	No mention found	No mention found
Garcia et al., 2018	News comment sentiment, regression	No mention found; ”high accuracy”	Comparable to polls (claimed by authors)	4 months pre-election
Jain and Kumar, 2017	Social media/news, Support Vector Machine (SVM)	Accuracy, F-measure (79.4% SVM)	Outperformed traditional surveys (as reported in the full text)	2 months pre-election, real-time
Huang, 2017	Social media, opinion extraction	Vote share difference (17.74%)	No comparison to polls	No mention found
Sanger, 2019	Econometric/textual analysis	No mention found	No mention found	Real-time data, various elections

Approach:

- Social media sentiment analysis:4 studies (including Twitter, news comments, or opinion extraction).

- News salience/Semantic Brand Score:1 study.
- Hybrid approach (social media and polls):1 study.
- Meta-analysis or extension methods:2 studies (Supervised Aggregated Sentiment Analysis, ReadMe extension, or meta-analysis of sentiment).
- Lexicon-based sentiment analysis:1 study.
- Econometric/textual analysis:1 study.

Performance vs Polls:

- Outperformed polls or traditional surveys:4 studies.
- Comparable to polls (claimed):1 study.
- Noteworthy correlation with mass surveys:1 study.
- No mention found:3 studies.
- No comparison to polls:1 study.

Accuracy Metric:

- Specified accuracy metric:4 studies (Mean Absolute Percentage Error/error margins, lower standard errors, accuracy/F-measure, or vote share difference).
- No mention found:6 studies.

Forecasting Window:

- Real-time or nowcasting approaches:3 studies.
- Retrospective (2011–2012 or 2011–2013):2 studies.
- Pre-election window (weekly/1 week, 4 months, 2 months):3 studies.
- No mention found:3 studies.

Contribution of Media Data to Hybrid Models

Several studies examined the integration of media data with traditional polls:

- Improved results with hybrid models:Tsakalidis et al. (2015) found that combining Twitter features with poll data in a multivariate time-series model outperformed using either source alone or commercial baselines (as reported in the abstract).
- Supplementing traditional surveys:Sanger (2019) and Ceron et al. (2013) highlighted the value of using social media data to supplement or consolidate traditional surveys, with evidence of improved correlation and explanatory power (as reported in the abstracts).
- Complementing polls with Semantic Brand Score:Colladon (2020) suggested that Semantic Brand Score can complement polls, especially when polling is restricted or less reliable, and that hybrid approaches may be optimal in practice.

Limitations:

- Direct comparisons limited:Not all studies directly compared hybrid and single-source models.
- Context-dependent improvement:The degree of improvement varied by context.
- Evidence base:The included studies support the utility of hybrid models, particularly for short-term forecasting and in environments where polling is limited or delayed.

Temporal Dynamics and Short-term Prediction Performance

Timing was a key factor in the studies reviewed:

- Peak accuracy close to election:Colladon (2020) found that Semantic Brand Score-based forecasts were most accurate one week before the election, with performance declining at longer lead times.
- Real-time and near-real-time data:Jain and Kumar (2017) and Sanger (2019) used real-time or near-real-time data, with evidence that accuracy improved as the election approached and as the volume of relevant media data increased.
- Longer pre-election window:Garcia et al. (2018) collected data up to four months pre-election but did not report on temporal changes in accuracy.

Synthesis:

- High-frequency media data:The available studies suggest that high-frequency media data is most valuable in the final days before an election.
- Alignment with current context:This aligns with the context of forecasting five days before the Montréal municipal election, supporting the use of rapid, high-resolution media salience measures for short-term electoral forecasting.

References

A. C. Garcia, William Silva, and L. Correia. "The PredNews Forecasting Model." *Digital Government Research*, 2018.

A. Ceron, L. Curini, and S. Iacus. "Politics and Big Data : Nowcasting and Forecasting Elections with Social Media," 2016.

———. "Using Social Media to Forecast Electoral Results: A Review of the State of the Art," 2013.

A. Ceron, L. Curini, S. Iacus, and G. Porro. "Every Tweet Counts? How Sentiment Analysis of Social Media Can Improve Our Knowledge of Citizens' Political Preferences with an Application to Italy and France." *New Media & Society*, 2013.

A. F. Colladon. "Forecasting Election Results by Studying Brand Importance in Online News." *arXiv.org*, 2020.

Adam Tsakalidis, S. Papadopoulos, A. Cristea, and Y. Kompatsiaris. "Predicting Elections for Multiple Countries Using Twitter and Polls." *IEEE Intelligent Systems*, 2015.

D. Duval, and F. Pétry. "L'analyse Automatisée Du Ton Médiatique : Construction Et Utilisation de La Version Française Du Lexicoder Sentiment Dictionary." *Canadian Journal of Political Science/Revue Canadienne de Science Politique*, 2016.

Jia-Yen Huang. "Web Mining for the Mayoral Election Prediction in Taiwan." *Aslib Journal of Information Management*, 2017.

V. Jain, and Shishir Kumar. "Towards Prediction of Election Outcomes Using Social Media," 2017.

William Sanger. "Science Des Données Et Politique : Quatre Essais Pour Comprendre Les Processus Démocratiques," 2019.